

"RPI_Robot_HAT" is a raspberry PI HAT designed for small/medium size robots. It has an IMU, Dynamixel connectors and audio. Its dimensions fits a raspberry PI zero

Schematic

PCB01186
ASE01187
Logo Pollen

Fiducials

Cabled buses on Raspberry PIs
- **PI zero 2W:**
  - **I2C0 (pin 27/28 ID_SD/C): reserved, but can be on J6 with 0R**
  - **I2C1 (IO 2/ 3 - pin 3/ 5 SDA/SCL): IMU + Audio + Qwiic J5**
- **PI 4:**
  - **same I2C0/1**
  - **I2C3 (IO 4/ 5 - pin 7/29 SDA/SCL): Qwiic J6**
  - **then choose between:**
  - **I2C4 (IO 8/ 9 - pin 24/21 SDA/SCL): Qwiic J7**
  - **I2C5 (IO 10/11 - pin 19/23 SDA/SCL): Qwiic J8**
- **PI 5:**
  - **same I2C0/1**
  - **I2C2 (IO 2/ 3 - pin 3/ 5 SDA/SCL): Qwiic J5**

Cabled GPIOs on the 40-way Header:
- GPIO00/01: HAT identification over I2C (reserved)
- GPIO02: IMU + Audio + Stemma/QT - SDA
- GPIO03: IMU + Audio + Stemma/QT - SCL
- GPIO04: Stemma/QT - SDA
- GPIO05: Stemma/QT - SCL
- GPIO06: Switch off detection
- GPIO07
- GPIO08: Stemma/QT - SDA
- GPIO09: Stemma/QT - SCL
- GPIO10: Stemma/QT - SDA
- GPIO11: Stemma/QT - SCL
- GPIO12
- GPIO13:
- GPIO14: Dynamixel - UART_Tx
- GPIO15: Dynamixel - UART_Rx
- GPIO16
- GPIO17
- GPIO18: Audio, PCM Clk
- GPIO19: Audio, PCM FS
- GPIO20: Audio, Din
- GPIO21: Audio, Dout
- GPIO22: IMU_IT
- GPIO23
- GPIO24
- GPIO25
- GPIO26
- GPIO27

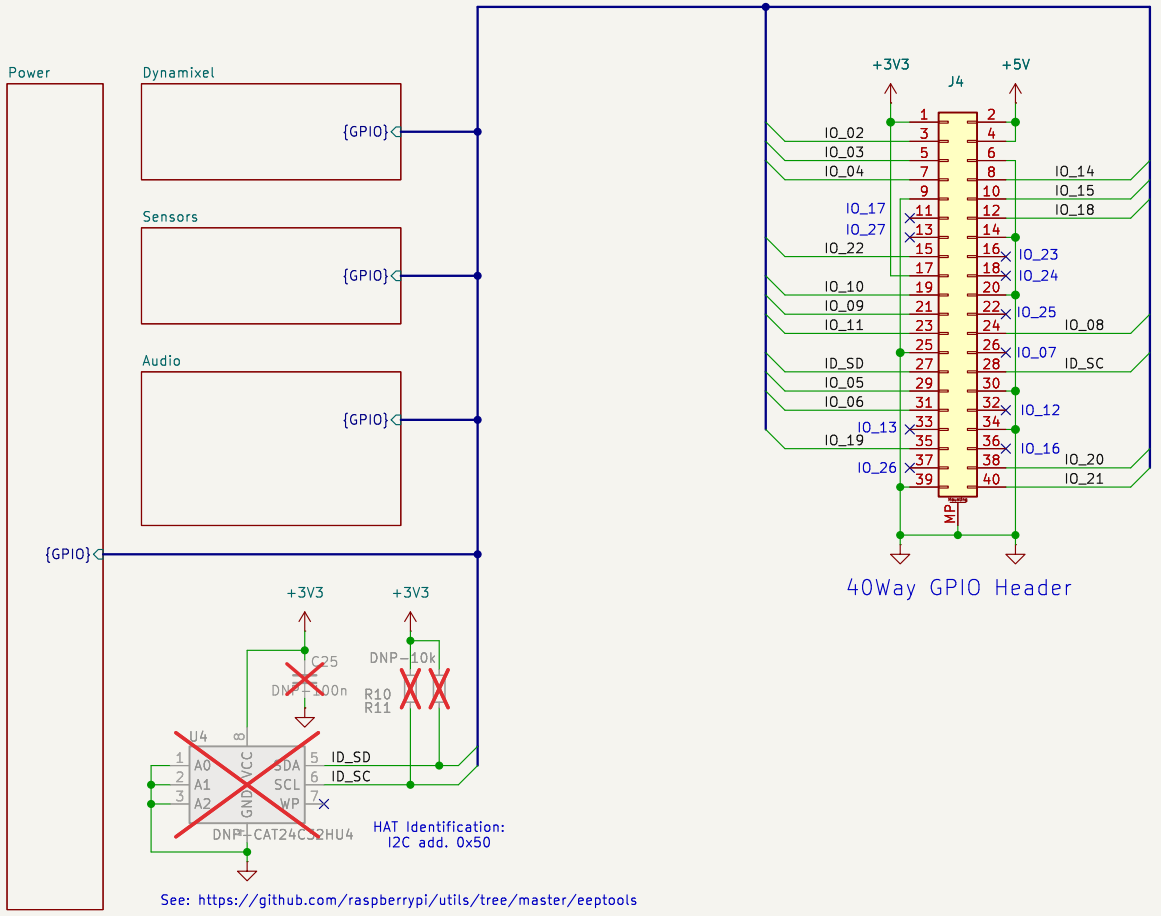
POLLEN ROBOTICS

Pollen Robotics / Hugging Face

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File: elec_RPI_Robot_HAT.kicad_sch

Title: elec RPI Robot HAT

Size: A4	Date: 2026-11-09	**Rev: D1-M1**
KiCad E.D.A. 9.0.7		Id: 1/6



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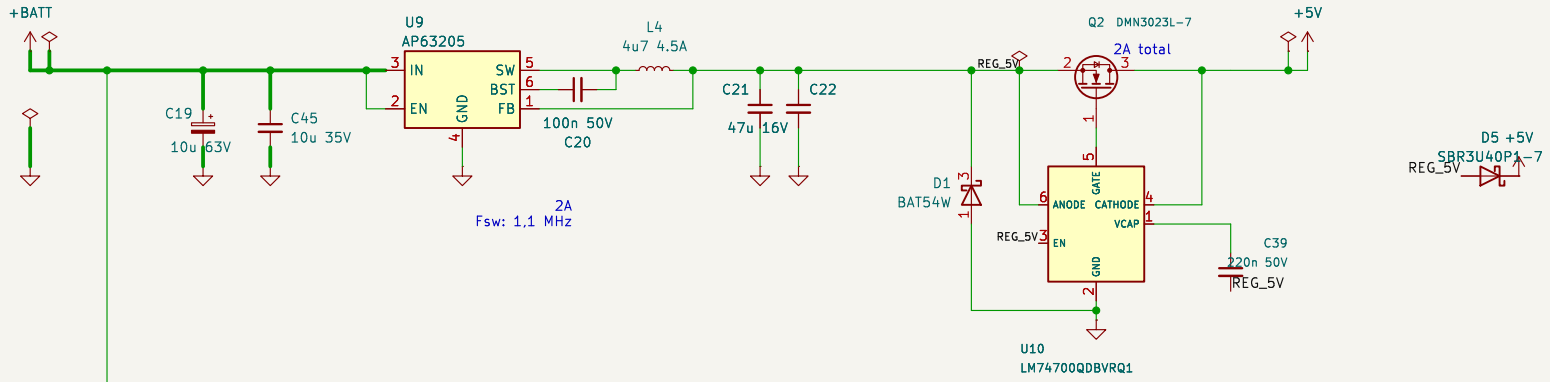
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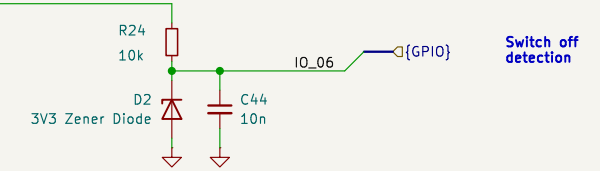
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Power

2S battery

6.0–8.4V M1 input
See ORDERING.md





Model	Idle (W)	Charge max. (W)	"Usage moyen"* (W)
Raspberry Pi 5	2,4 – 3,3	9 – 10	4,0
Raspberry Pi 4 B	2,5 – 3,4	6 – 7,6	3,5
Raspberry Pi Zero 2 W	0,6	1,5	0,8

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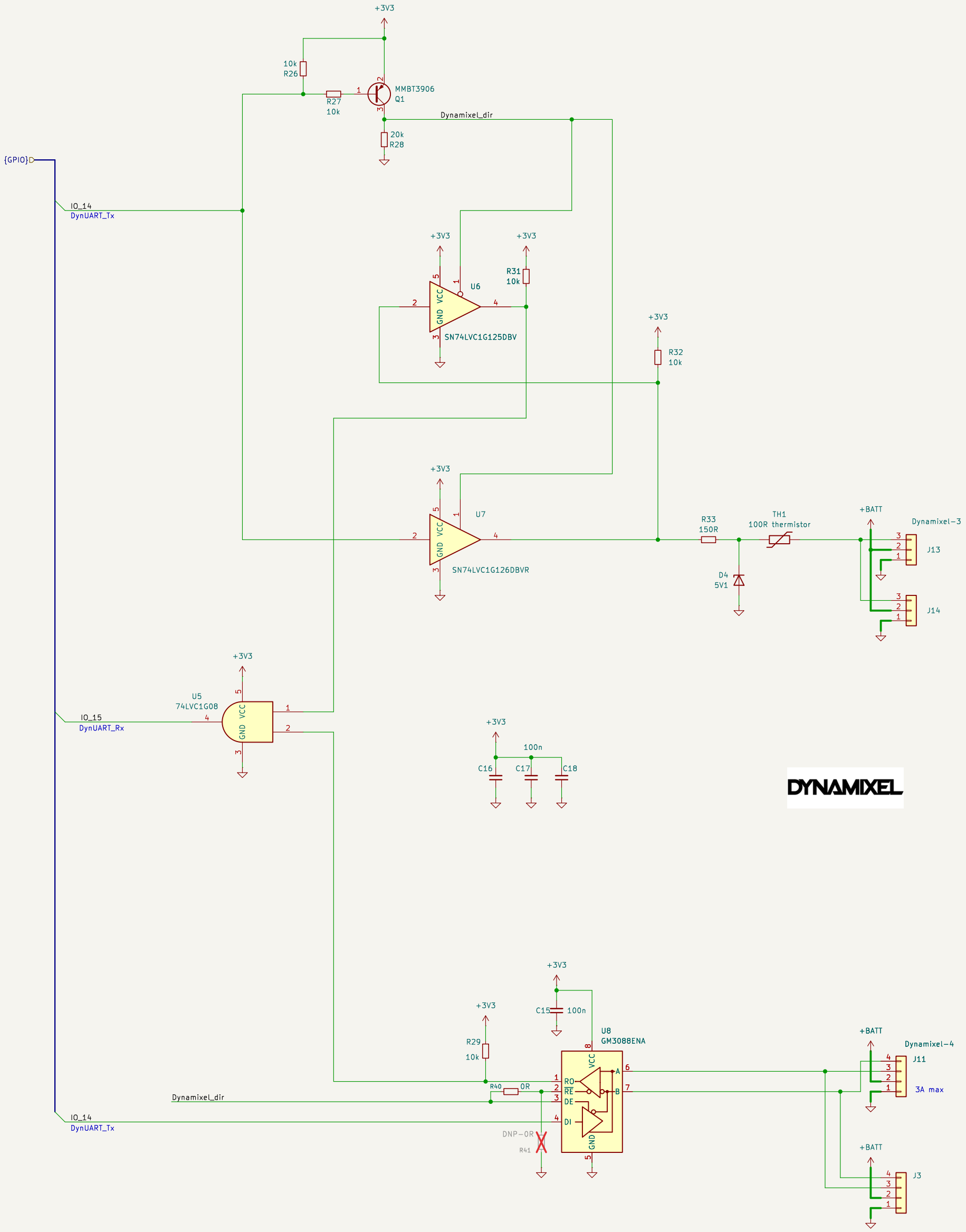
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Title: elec RPI Robot HAT

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Dynamixel

See: https://emanual.robotis.com/docs/en/parts/interface/dynamixel_shield/
as a reference.



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Title: elec RPI Robot HAT		
Size: A3	Date: 2026-11-09	**Rev: D1-M1**
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Sensors

{GPIO}

+3V3

10k R12 R13

SDA IO_02
IO_03 SCL

J5 JST SH

+3V3

10k R20 R21

SDA IO_04 R18 0R
IO_05 SCL R19 0R

ID_SD R17 DNP-0R
ID_SC R16 DNP-0R

J6 JST SH

+3V3

10k R34 R35

SDA IO_08
IO_09 SCL

J7 JST SH

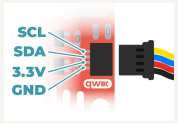
+3V3

10k R38 R39

SDA IO_10
IO_11 SCL

J8 JST SH

STEMMA QT 1mm
3V3 Default



IMU

{GPIO}

+3V3

R9 0R

C23 C24
100n 16V

SDA IO_02
SCL IO_03

R30 0R
R14 0R
R15 0R

SDO_Aux
SDO/SA0
SDX
SCX
OCS_Aux
SDA
SCL
CS
VDDIO
VDD
GND
INT1
INT2

R25 0R IO_22 {GPIO}
TP5

U12
LSM6DSV16XTR

I2C Addr: 0x6A

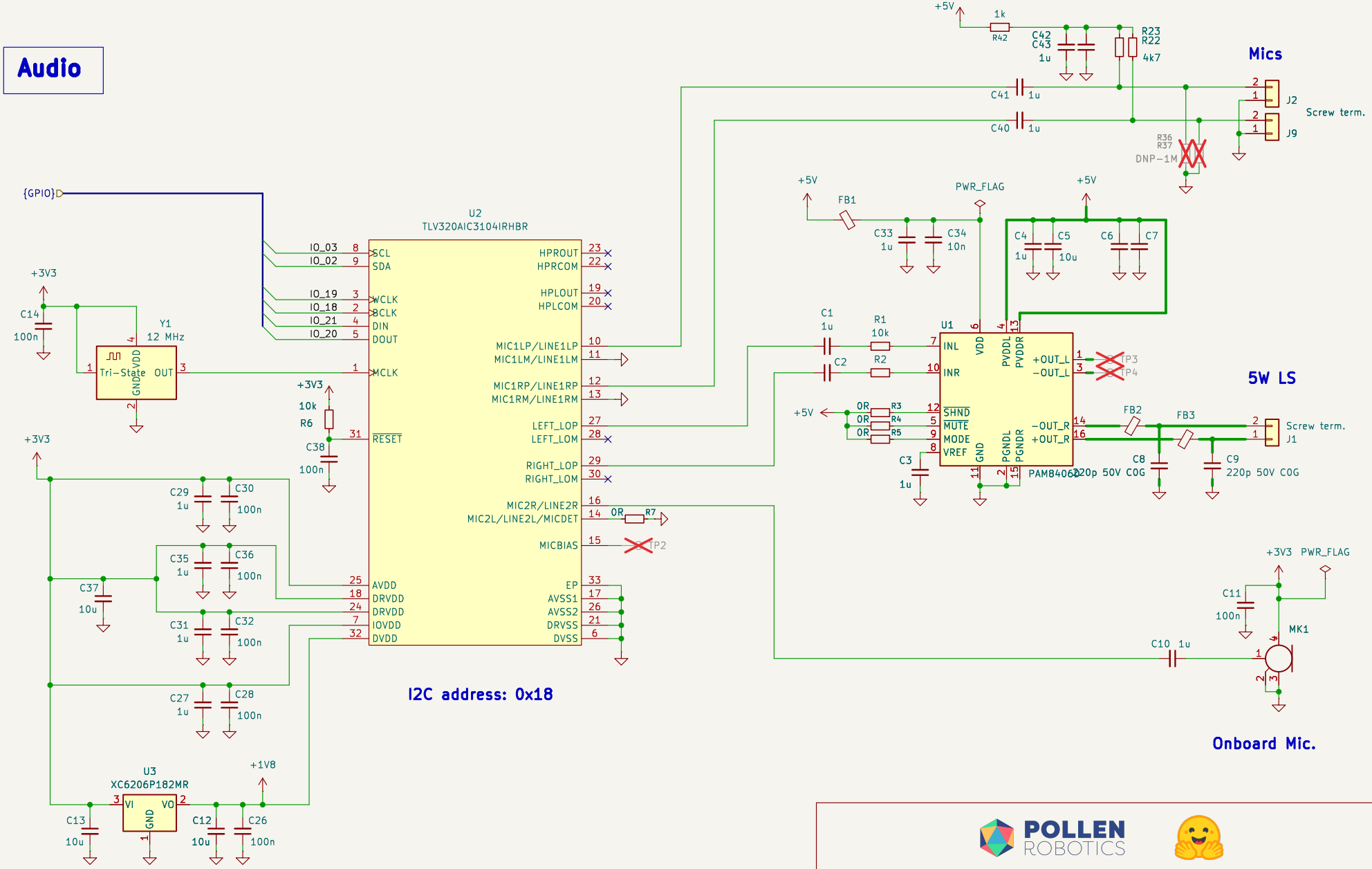
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Audio



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